Cool Quantitative Fewnomial Facts and More

★ Fewnomial Upper Bounds Over $\mathbb{F}_q$: Let $f\in\mathbb{F}_q[x]$ be a univariate t-nomial with exponents $0=:e_1<e_2<\cdots<e_t$ and set $\delta(f):=\gcd(e_1,\ldots,e_t,q-1)$, $Z_q^*(f):=\left\{x\in\mathbb{F}_q^*\ \middle|\ f(x)=0\right\}$, $R_q^*(f):=\#Z_q^*(f)$, and let $C_q(f)$ be the maximum cardinality of any coset (of any subgroup of $\mathbb{F}_q^*$) on which f vanishes. Then...

0. [Karinski-Shparlinski, 1996] $\boxed{R_q^*(f)\leq\left(1-\frac{1}{t}\right)(q-1)}$ and, when $q=1 \bmod t$, this is attained by $\frac{x^{q-1}-1}{x^{(q-1)/t}-1}$ (but note that this example has $\delta=\frac{q-1}{t}$ which is often >1).

1. [Kelley, Owen, 2016, Thm. 1.2] For $t=3$ we have $\boxed{R_q^*(f)\leq\delta(f)\left\lfloor\frac{1}{2}+\sqrt{\frac{q-1}{\delta(f)}}\right\rfloor}$ and, if $\delta=1$ and q is a perfect square as well, then $\boxed{R_q^*(f)=\sqrt{q}}$.

2. [Kelley, 2016, Thms. 2.2 & 2.3] For any $t\geq 2$ we have $\boxed{R_q^*(f)\leq 2(q-1)^{1-\frac{1}{t-1}}C_q(f)^{\frac{1}{t-1}}}$, $C_q(f)\geq\delta(f)$, and $C_q(f)$ is no greater than the *largest divisor* k *of* $q-1$ *such that,* for all i, there is a $j\neq i$ with $k|(e_i-e_j)$.

★ Fewnomial Lower Bounds Over $\mathbb{F}_q$: [Cheng, Gao, Rojas, Wan, 2017]

3. $\boxed{f(x):=1+x+x^{p^u}+\cdots+x^{p^{(t-2)u}}\text{ has exactly }q^{\frac{t-2}{t-1}}\text{ roots in }\mathbb{F}_q^*,}$ where $q:=p^{(t-1)u}$. Note that $\delta(f)=C_q(f)=1$ here.

4. $\boxed{f(x):=1+x+x^{1+p^u}+\cdots+x^{1+p^u+\cdots+p^{(t-2)u}}\text{ has exactly } q^{\frac{t-2}{t}}+\cdots+q^{\frac{1}{t}}+1\text{ roots in }\mathbb{F}_q^*,\text{ where }q:=p^{tu}.}$ Note that $\delta(f)=1$ and $C_q(f)\leq\lfloor t/2\rfloor$ here.

5. The *prime* field case is far more subtle and, for $t=3$ and $\delta(f)=1$, it is tempting to conjecture to conjecture an $o(\sqrt{p})$ bound like $O(\log p)$ or even $O\left(\frac{\log p}{\log\log p}\right)$. There is evidence from the first 25450 primes after 3: The smallest primes p_n yielding a trinomial f_n with exactly n roots in $\mathbb{F}_{p_n}$ (and $\delta(f_n)=1$), for $n\in\{2,\ldots,16\}$ are:

n	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
p_n	5	11	23	47	151	173	349	619	1201	2753	4801	10867	16633	71237	8581

$\boxed{\text{No }p\leq 292837\text{ yields an }f\in\mathbb{F}_p[x]\text{ with }\delta(f)=1\text{ and 17 roots in }\mathbb{F}_p^*,}$ thanks to supercomputer searches by Zander Kelley (before 2017) and Viviana Marquez (as of May 2020). In particular, upper bounds of $3.893\frac{\log p}{\log\log p}$ and $1.767\log p$ appear to hold so far.

6. The underlying trinomials (of minimal degree and minimal maximal coefficient absolute value, normalized so that 1 is a root) are:

$f_2:=1+x-2x^2$, $f_3:=1-3x+2x^3$, $f_4:=-2+x+x^4$, $f_5:=1+4x-5x^8$, $f_6:=1+24x-25x^{33}$, $f_7:=-2+x+x^{34}$, $f_8:=1+23x-24x^{21}$, $f_9:=-71+70x+x^{184}$, $f_{10}:=1+5x-6x^{152}$, $f_{11}:=-797+796x+x^{67}$, $f_{12}:=-82+81x+x^{1318}$, $f_{13}:=-1226+1225x+x^{225}$, $f_{14}:=-39+38x+x^{2264}$, $f_{15}:=29574-29573x-x^{27103}$, and $f_{16}:=-364+363x+x^{2729}$.

7. For any $n\geq 2$, there is a prime $p_n\in\mathbb{N}$ such that the trinomial $\boxed{x^n-x-1\text{ has exactly }n\text{ roots in }\mathbb{F}_{p_n}\text{ and }p_n=e^{(2n+1+o(1))\log n}}$ if GRH holds (and $p_n=e^{e^{n+\frac{1}{2}+o(1))\log n}}$ unconditionally). So there are trinomials in $\mathbb{F}_p[x]$ with $\Omega\left(\frac{\log p}{\log\log p}\right)$ roots in $\mathbb{F}_p^*$ if GRH holds.

★ Upper Bounds Over $\mathbb{R}$: $(n+k)$-nomial $n\times n$ Systems and Beyond: Let $Y_{\mathbb{R}}(n,k)$ denote the maximal (finite) number of non-degenerate roots of a real $(n+k)$-nomial $n\times n$ system in $\mathbb{R}_+^n$. Then...

8. $\boxed{Y_{\mathbb{R}}(1,k)=k}$ [Descartes, ≤1637 & Wang, 2004], $\boxed{Y_{\mathbb{R}}(n,1)=1}$ [Folklore, ≤1990s], and the same bounds holds if one counts isolated roots.

9. [Bertrand, Bihan, Sottile, 2005] (see also [Li, Rojas, Wang, 2003]) $\boxed{Y_{\mathbb{R}}(n,2)=n+1}$.

Now consider $h(x):=\sum_{i=1}^t c_ix^{a_i}(1-x)^{b_i}$ for $a_i,b_i,c_i\in\mathbb{R}$ and $c_i\neq 0$ for all i. Then...

10. [El Hilani, 2015] The # of roots of h in the open interval $(0,1)$ is $\boxed{\leq 3\cdot 2^{t-2}-1}$...

11. [Koiran, Portier, Tavenas, 2015] ...and $\boxed{\leq\frac{2}{3}t^3+5t}$ (and El Hilani's bound is better for $t\in\{2,\ldots,9\}$).

12. [Bihan, Sottile, 2007] Suppose $\beta_{\ell,r}\in\mathbb{R}$ for all $(\ell,r)\in\{1,\ldots,m+j\}\times\{0,\ldots,j\}$, $[e_{i,\ell}]\in\mathbb{R}^{j\times(m+j)}$, and $G:=G(\lambda_1,\ldots,\lambda_j)$ is defined to be

$$\begin{bmatrix}\prod_{\ell=1}^{m+j}(\beta_{\ell,0}+\beta_{\ell,1}\lambda_1+\cdots+\beta_{\ell,j}\lambda_j)^{e_{1,\ell}}\\ \vdots\\ \prod_{\ell=1}^{m+j}(\beta_{\ell,0}+\beta_{\ell,1}\lambda_1+\cdots+\beta_{\ell,j}\lambda_j)^{e_{j,\ell}}\end{bmatrix}.$$

We call such a system a $j\times j$ *Gale dual system with* $m+j$ *factors.* Then

$\boxed{G\text{ has }<\frac{e^2+3}{4}(\sqrt{2}^{j-1}m)^j\text{ non-degenerate roots }(\lambda_1,\ldots,\lambda_j)\text{ in }\Delta:=\{\beta_{\ell,0}+\beta_{\ell,1}\lambda_1+\cdots+\beta_{\ell,j}\lambda_j>0\text{ for all }\ell\in\{1,\ldots,m+j\}\}.}$

13. Corollary: Since any real $(n+k)$-nomial $n\times n$ system F is equivalent (via linear algebra and multiplying/dividing equations) to a Gale dual system with $j=k-1$ and $m=n$, such an F has $\boxed{<\frac{e^2+3}{4}2^{(k-1)(k-2)/2}n^{k-1}\text{ non-degenerate roots in }\mathbb{R}_+^n.}$ $\left(\frac{e^2+3}{4}=2.597\ldots\right)$

★ Fewnomial Lower Bounds Over $\mathbb{R}$:

14. $\boxed{x\left(x^2-1^2\right)\cdots\left(x^2-(t-1)^2\right)}$ has exactly $\boxed{t}$ (resp. $\boxed{2t-1}$) roots in $\mathbb{R}_+$ (resp. $\mathbb{R}$).

15. [Phillipson, Rojas, 2013] A family of $(n+2)$-nomial $n\times n$ systems attaining $n+1$ positive roots is

$$\boxed{\left(x_1x_2-\tfrac{1}{4}\left(1+4x_1^2\right),x_2x_3-\left(1+\tfrac{x_1^2}{4}\right),x_3x_4-\left(1+\tfrac{x_1^2}{4^3}\right),\ldots,x_{n-1}x_n-\left(1+\tfrac{x_1^2}{4^{2n-5}}\right),x_n-\left(1+\tfrac{x_1^2}{4^{2n-3}}\right)\right).}$$

16. The Gale dual form of the preceding system is

$$\boxed{u\left(1+\tfrac{u}{4}\right)^2\left(1+\tfrac{u}{4^5}\right)^2\cdots\left(1+\tfrac{u}{4^{4\lfloor n/2\rfloor-3}}\right)^2-\tfrac{1}{4^2}\left(1+4u\right)^2\left(1+\tfrac{u}{4^3}\right)^2\cdots\left(1+\tfrac{u}{4^{4\lceil n/2\rceil-5}}\right).}$$

17. [Bihan, Rojas, Sottile, 2007] $\boxed{Y_{\mathbb{R}}(n,k)\geq\left\lfloor\frac{n+k-1}{\min\{n,k-1\}}\right\rfloor^{\min\{n,k-1\}}}$ (and a slightly larger, more intricate bound [PR13, Thm. 1.5] also holds).

★ Fewnomial Upper Bounds Over $\mathbb{Q}_p$ and $\mathbb{F}_q((t))$: For any $L\in\{\mathbb{Q}_p,\mathbb{F}_q((t))\}$, let $Y_L(n,k)$ denote the maximal (finite) number of roots in $(L^*)^n$, with all coordinates having most significant digit 1, of a p-adic rational $(n+k)$-nomial $n\times n$ system. Then (see also [PR13])...

18. Since the group of roots of unity in $\mathbb{Q}_p^*$ is cyclic of order $\max\{2,p-1\}$, Gaussian Elimination $\Longrightarrow$ $\boxed{Y_{\mathbb{Q}_2}(n,1)=2^n}$ and $\boxed{Y_{\mathbb{Q}_p}(n,1)=1\text{ for all }p\geq 3}$.

19. $\boxed{Y_{\mathbb{Q}_2}(1,2)=6}$ [Lenstra, 1997], $\boxed{Y_{\mathbb{Q}_3}(1,2)=4}$ [Rojas, Zhu, 2020], and $\boxed{Y_{\mathbb{Q}_p}(1,2)=3}$ for all $p\geq 5$ [Avendaño, Krick, 2011] (counting multiplicities).

20. [Poonen, 1998] $\boxed{Y_{\mathbb{F}_q((t))}(1,k)=\frac{q^k-1}{q-1}}$.

21. $\boxed{Y_{\mathbb{Q}_p}(1,k)\leq k^2-k+1\text{ for }p>1+k}$ [Avendaño, Krick, 2011] and $\boxed{Y_{\mathbb{Q}_p}(1,k)\leq\frac{e}{e-1}k^2\left(1+\frac{\log(k/\log p)}{\log p}\right)\text{ for all }p}$ [Lenstra, 1997]. $\left(\frac{e}{e-1}=1.5819\ldots\right)$

22. [Rojas, 2004] If the underlying tropical varieties over $\mathbb{C}_p$ intersect transversally then $\boxed{Y_{\mathbb{Q}_p}(n,k)\leq u(n,k)\left\lceil\left[\frac{e}{e-1}kn\left(1+\frac{\log(k/\log p)}{\log p}\right)\right]^n\right\rceil}$ (counting multiplicities), where $u(n,k)$ is k, $\max\{1,9(k-1)^2\}$, or $(k(k+1)/2)^n$, according as n is 1, 2, or ≥ 3.

★ Fewnomial Lower Bounds Over $\mathbb{Q}_p$ (and more):

23. $\boxed{\left(x_1^{\max\{2,p-1\}} - 1, \ldots, x_n^{\max\{2,p-1\}} - 1\right)}$ attains the upper bound $Y_{\mathbb{Q}_p}(n,1)$, and has maximally many roots in $(\mathbb{Q}_p^*)^n$ as well.
24. $\boxed{(x_1^2-1)(x_1^2-4)\cdots(x_1^2-4^{k-1})}$ yields $Y_{\mathbb{Q}_2}(1,k)\geq 2k$.
25. $\boxed{3x_1^{10}+x_1^2-4}$ [Lenstra, 1997] and $\boxed{x_1^{10}+11x_1^2-12}$ [Rojas, 2021] each yield $Y_{\mathbb{Q}_2}(1,2)\geq 6$.
26. [Goldstein, Rojas, 2021] $\boxed{\frac{p^{p+3}(p-1)}{2} + p^{p-1} - \left(1+p^{p-1}\right)x + x^{1+p^{p-1}}}$ yields $Y_{\mathbb{Q}_3}(1,2)\geq 4$ and $Y_{\mathbb{Q}_p}(1,2)\geq 3$ for all $p\geq 5$ (and 8 roots in $\mathbb{Q}_3$ and $3(p-1)$ roots in $\mathbb{Q}_p$ for all $p\geq 5$).
27. [Avendaño, Krick, 2011] $Y_{\mathbb{Q}_p}(1,k)\geq 2k-1$ for all $p\geq 3$.
28. [Poonen, 1998] $\boxed{\prod_{z_1,\ldots,z_{k-1}\in\mathbb{F}_q}(x_1-z_1-z_2t-\cdots-z_{k-1}t^{k-1})}$ yields $Y_{\mathbb{F}_q((t))}(1,k)\geq \frac{q^k-1}{q-1}$.
29. [Phillipson, Rojas, 2013] The system

$$\boxed{\begin{array}{l}\left(x_1x_2 - p\left(1+\frac{x_1^2}{p}\right), x_2x_3-\left(1+px_1^2\right), x_3x_4-\left(1+p^3x_1^2\right),\right.\\ \left.\quad\ldots, x_{n-1}x_n-\left(1+p^{2n-5}x_1^2\right), x_n-\left(1+p^{2n-3}x_1^2\right)\right)\end{array}}$$

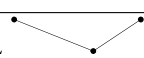
yields $Y_{\mathbb{Q}_p}(n,2)\geq n+1$. Furthermore, replacing p by t yields $Y_{\mathbb{F}_q((t))}(n,2)\geq n+1$ as well.

★ $\mathcal{A}$-hypergeometric Series for Roots of Trinomials in $\mathbb{C}$ and $\mathbb{C}_p$

Consider $f(x_1):=1+cx_1^m+x_1^n\in\mathbb{C}[x_1]$ where $c\neq 0$, $0<m<n$, and $\gcd(m,n)=1$. Series for the complex roots of f date back to 1757 work of Johann Lambert for the special case $m=1$. Several authors have since extended these series in various directions. Passare and Tsikh's 2004 paper (in a Springer volume on the legacy of Niels Henrik Abel) was the most useful for this summary. The union of the domains of convergence of our series will turn out to be all c with $|c|$ distinct from $r_{m,n}:=\frac{n}{m^{\frac{m}{n}}(n-m)^{\frac{n-m}{n}}}$ (>1). The three families of series are the following:

$|c|<r_{m,n}\Longrightarrow$ coarsest lower hull : Let v_n be any nth root of -1. There are then exactly n series (giving roots in the annulus $\frac{1}{2}<|x_1|<2$), each converging provided $|c|<r_{m,n}$:

$$\boxed{x_{\mathrm{mid}}(c) = v_n\left[1+\sum_{k=1}^{\infty}\left(\frac{v_n^{mk}}{kn^k}\cdot\prod_{j=1}^{k-1}\frac{1+km-jn}{j}\right)c^k\right].}$$

$|c|>r_{m,n}\Longrightarrow$ finest lower hull : Let v_m and v_{n-m} respectively denote any mth and $(n-m)$th roots of -1. There are then exactly m series of the form:

$$\boxed{x_{\mathrm{low}}(c) = \frac{v_m}{|c|^{1/m}}\left[1+\sum_{k=1}^{\infty}\left(\frac{v_m^{nk}}{km^k}\cdot\prod_{j=1}^{k-1}\frac{1+kn-jm}{j}\right)\left(\frac{1}{|c|^{n/m}}\right)^k\right]}$$

(yielding roots with norm within a factor of 2 of $|c|^{-1/m}$), and exactly $n-m$ series of the form:

$$\boxed{x_{\mathrm{hi}}(c) = v_{n-m}|c|^{\frac{1}{n-m}}\left[1-\sum_{k=1}^{\infty}\left(\frac{v_{n-m}^{-nk}}{k(n-m)^k}\cdot\prod_{j=1}^{k-1}\frac{km+j(n-m)-1}{j}\right)\left(\frac{1}{|c|^{\frac{n}{n-m}}}\right)^k\right]}$$

(yielding roots with norm within a factor of 2 of $|c|^{1/(n-m)}$). The last two families of series, $x_{\mathrm{low}}(c)$ and $x_{\mathrm{hi}}(c)$, converge when $|c|>r_{m,n}$.

Classical Facts:

Recall that $\boxed{\Gamma(z):=\int_0^\infty x^{z-1}e^{-x}dx}$ and $\boxed{\beta(x,y):=\int_0^1 t^{x-1}(1-t)^{y-1}dt}$ satisfy $\beta(x,y)=\frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$. In particular, Γ is meromorphic, with (simple) poles at $0,-1,-2,\ldots$ and an essential singularity at ∞; and $1/\Gamma(z)$ is analytic. Also, $\boxed{\Gamma(n)=(n-1)!}$ for all $n\in\mathbb{N}$, $\Gamma(z+1)=z\Gamma(z)$ for all $z\in\mathbb{C}\setminus(-\mathbb{N}\cup\{0\})$, $\Gamma(1/2)=\sqrt{\pi}$, and $\Gamma(-1/2)=-2\sqrt{\pi}$.
$\boxed{e^{7/8}\sqrt{n}(n/e)^n<\Gamma(n+1)<e\sqrt{n}(n/e)^n \text{ for all } n\geq 2,}$ and $\Gamma(n+1)\sim\sqrt{2\pi n}(n/e)^n$.

$(e=2.718281828\ldots,\ \sqrt{2\pi}=2.506628274\ldots,\text{ and } e^{7/8}=2.398875294\ldots)$

Furthermore, $\Gamma(1-z)\Gamma(z)=\frac{\pi}{\sin(\pi z)}$ for all $z\in\mathbb{C}\setminus\mathbb{Z}$ and $\sqrt{\pi}=1.772453851\ldots$

★ *If the Riemann Hypothesis is true then...*

$\boxed{\left|\pi(x)-\int_2^x\frac{dt}{\log t}\right|<\frac{\sqrt{x}\log x}{8\pi}}$ for all $x\geq 2657$ [Schoenfeld, 1976] and, for all $x\geq 2$ there is a prime p satisfying $\boxed{x-\frac{4\sqrt{x}\log x}{\pi}<p\leq x}$ [Dudek, 2014].

★ *If the Generalized Riemann Hypothesis is true then...*

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★ Quadratic Reciprocity: Suppose n is odd and factors as $p_1^{b_1}\cdots p_r^{b_r}$; $a,b\in\mathbb{Z}$; and we define $\left(\frac{a}{n}\right):=\left(\frac{a}{p_1}\right)^{b_1}\cdots\left(\frac{a}{p_r}\right)^{b_r}$ and, when n is prime, define $\left(\frac{a}{n}\right)$ to be $a^{(n-1)/2}$ or 0 according as $\gcd(a,n)=1$ or not. Then, when n prime, a has a well-defined square root in $\mathbb{F}_n\Longleftrightarrow\left(\frac{a}{n}\right):=1$. More generally...

(1) $[a=b \bmod n \text{ and } \gcd(a,n)=1]\Longrightarrow\left(\frac{a}{n}\right)=\left(\frac{b}{n}\right)$.

(2) $\gcd(ab,n)=1\Longrightarrow\left(\frac{ab}{n}\right)=\left(\frac{a}{n}\right)\left(\frac{b}{n}\right)$. (3) $\left(\frac{-1}{n}\right)=(-1)^{(n-1)/2}$.

(4) $\left(\frac{2}{n}\right)$ is 1 or -1 according as $n\in\{\pm 1\}$ or $n\in\{\pm 3\}$.

$\boxed{5}$ a odd $\Longrightarrow\left(\frac{a}{n}\right)$ is $-\left(\frac{n}{a}\right)$ if $a=n=-1 \bmod 4$, and $\left(\frac{n}{a}\right)$ otherwise.

★ Effective Chebotarev: Let K be an algebraic number field and L a normal extension with Galois group $G=G(L/K)$, d_L and d_K the respective absolute values of the discriminants of L and K, and $n_L:=[L:\mathbb{Q}]$ and $n_K:=[K:\mathbb{Q}]$. Let $\mathcal{P}$ denote a prime ideal of K and P a prime ideal of L. If $\mathcal{P}$ is unramified in L then we use the Artin symbol $\left[\frac{L/K}{\mathcal{P}}\right]$ to denote the conjugacy class of Frobenius automorphisms corresponding to prime ideals $P|\mathcal{P}$. For each conjugacy class C of G we define

$\pi_C(x,L/K):=\#\left\{\mathcal{P}\mid\mathcal{P}\text{ unramified in } L, \left[\frac{L/K}{\mathcal{P}}\right]=C, N_{K/\mathbb{Q}}\mathcal{P}\leq x\right\}$.

Prime Ideal Theorem *($L=K$, with K not necessarily normal over $\mathbb{Q}$, and thus $G=\{e\}=C$) The number of prime ideals of K with norm $\leq x$ is asymptotic to* $\mathrm{Li}(x):=\int_2^\infty\frac{dt}{\log t}$.

Explicit Effective Chebotarev Theorem [Winckler, 2013] *If the Generalized Riemann Hypothesis is true then, for all $x\geq 2$, we have*

$$\left|\pi_C(x)-\frac{\#C}{\#G}\mathrm{Li}(x)\right|\leq\frac{\#C}{\#G}\sqrt{x}\left[\left(32+\frac{181}{\log x}\right)\log(d_L)+\left(28\log(x)+330+\frac{1655}{\log x}\right)n_L\right].$$

Unconditionally, for all $x\geq e^{8n_L(\log(150867d_L^{44/5}))^2}$, we have

$$\left|\pi_C(x)-\frac{\#C}{\#G}\mathrm{Li}(x)\right|\leq\frac{\#C}{\#G}\mathrm{Li}(x^\beta)+C_0xe^{-\frac{1}{99}\left(\frac{\log x}{n_L}\right)^{1/2}},$$

where $C_0=783846699796966<7.84\times 10^{14}$ and β is the largest real zero of ζ_L (currently known to be at least $1-\frac{1}{4\log d_L}$).

★ Arithmetic ζ Functions *For a scheme X over $\mathbb{F}_q$ we define the* arithmetic ζ function $\zeta_X(s):=\prod_x\frac{1}{1-\frac{1}{N(x)^s}}$ *where the product is over all closed points x of X and $N(x)$ is the cardinality of the residue field of x. (Equivalently, the product is over all points with $N(x)<\infty$.) A closely related function is $Z_X(t):=e^{\sum_{n=1}^\infty\frac{\#X(\mathbb{F}_{q^n})}{n}t^n}$. In particular, by Grothendieck's trace formula, $\zeta_X(s)=Z_X(q^{-s})$.*